

## Data communications (Lab 5): Unipolar NRZ Encoding

Objective: To generate and analyze a Polar NRZ-I (Non-Return-to-Zero Inverted) encoded signal using MATLAB from a given binary data sequence.

### Theory

Polar NRZ-I is a line coding technique used in digital communication where binary data is represented using two voltage levels (+1 and -1).

In Polar NRZ-I encoding:

- Binary 1 → change (transition) in signal level
- Binary 0 → no change in signal level

Thus:

- The signal depends on the previous state
- A transition occurs only when the input bit is 1

### MATLAB Code

```
clc;
clear;
close all;

b = [0 1 0 0 1 1 1 0];
n = length(b);
t = 0:.001:n;

for i = 1:n
    if (b(i) == 0)
        b_p(i) = 1;
    else
        b_p(i) = -1;
    end
end

for j = 1:n
    bw(j*1000:(j+1)*1000) = b_p(j);
end

bw = bw(1000:end);

subplot(2,1,1)
```

```

plot(t,bw)
grid on;

lastbit = 1;

for i = 1:n
    if b(i) == 1
        x(i*1000:(i+1)*1000) = -lastbit;
        lastbit = -lastbit;
    else
        x(i*1000:(i+1)*1000) = lastbit;
    end
end

x = x(1000:end);

subplot(2,1,2)
plot(t,x)
grid on;

```

### Code Explanation

#### 1. Initialization

```

clc;
clear;
close all;

```

- `clc` → Clears command window
- `clear` → Removes all variables

#### 2. Define Binary Input

```

b = [1 0 1 1 1 1 0];
n = length(b);

```

- `b` contains the binary input sequence
- `n` gives the total number of bits

#### 3. Create Time Vector

```

t = 0:0.001:n;

```

- Creates the time axis from 0 to n
- Step size 0.001 gives a smooth plot

#### 4. Polar Representation (Reference Signal)

```

if (b(i) == 0)
    b_p(i) = 1;
else
    b_p(i) = -1;

```

Converts:

- $0 \rightarrow +1$
- $1 \rightarrow -1$

#### 5. Signal Expansion

```
bw(j*1000:(j+1)*1000) = b_p(j);
```

- Each bit is expanded over 1000 samples
- Creates rectangular waveform

#### 6. NRZ-I Encoding

```
lastbit = 1;
```

- Initial signal level is assumed +1

```
if b(i) == 1
```

```
    x = -lastbit;
```

```
    lastbit = -lastbit;
```

- If bit = 1  $\rightarrow$  signal level changes

```
else
```

```
    x = lastbit;
```

```
    If bit = 0  $\rightarrow$  signal remains same
```

#### 7. Plotting

```
subplot(2,1,1)
```

```
plot(t,bw)
```

```
subplot(2,1,2)
```

```
plot(t,x)
```

- First plot  $\rightarrow$  Polar NRZ-Level (reference)
- Second plot  $\rightarrow$  Polar NRZ-I signal

#### Result:

The MATLAB program generates the Polar NRZ-I encoded waveform for the given binary sequence along with its corresponding polar representation.

#### Observations:

- A transition occurs for every binary 1
- No transition occurs for binary 0
- Signal depends on previous level
- NRZ-I provides better synchronization than NRZ-Level

**Conclusion:** The experiment successfully demonstrated Polar NRZ-I encoding using MATLAB. The binary input sequence was converted into a waveform where transitions represent binary 1 and no transitions represent binary 0. The results validate the working principle of Polar NRZ-I line coding.

Assignment: Write a MATLAB program to generate the Polar NRZ-L encoded signal for a binary sequence given as input by the user.